

Introduction to Artificial Intelligence

Test Document for Document Intelligence and Question Extraction

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is the field of computer science concerned with building systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, recognizing patterns, understanding language, solving problems, and making decisions.

Modern AI systems commonly use machine learning, where algorithms learn patterns from data rather than relying only on explicitly programmed rules. Deep learning is a specialized approach based on multi-layer neural networks and has been widely applied to computer vision, speech recognition, natural language processing, and recommendation systems.

2. Machine Learning

Machine learning is generally divided into supervised learning, unsupervised learning, and reinforcement learning. In supervised learning, a model learns from labeled examples. In unsupervised learning, the system attempts to discover patterns or structures in unlabeled data. Reinforcement learning involves an agent interacting with an environment and learning through rewards and penalties.

Learning Type	Basic Idea	Example
Supervised	Learn from labeled data	Email spam classification
Unsupervised	Find patterns in unlabeled data	Customer clustering
Reinforcement	Learn through rewards	Game-playing agent

3. Natural Language Processing

Natural Language Processing (NLP) enables computers to process and analyze human language. Applications include machine translation, sentiment analysis, text classification, information retrieval, and conversational assistants. NLP systems may use statistical methods, neural networks, transformer architectures, or combinations of these techniques.

4. Responsible AI

Responsible AI focuses on developing and deploying AI systems in ways that consider reliability, fairness, privacy, transparency, security, and human oversight. Evaluation should consider not only model accuracy but also how a system behaves in realistic situations and how errors affect users.

5. Key Concepts

Training data: Data used to learn model parameters.

Inference: The process of using a trained model to produce an output.

Model: A computational representation that maps inputs to outputs.

Evaluation: Measuring system performance using appropriate metrics and test data.

6. Review Questions

A useful AI question should be grounded in the material, have a clear answer, and match an appropriate difficulty level. Questions may test recall, understanding, application, analysis, evaluation, or creation.